You are invited to the defense of

Scattering and Diffraction of Acoustic Waves in Three Problems with Broken Symmetry

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Thursday, March 12, 2026

1:00–2:00pm Central

Engineering Teaching Center, Room 4.120

`https://utexas.zoom.us/j/89513470060`

Abstract

The effects of asymmetry in acoustic scattering and diffraction phenomena are investigated across the frequency spectrum, simplifying the modeling of waves used for particle manipulation, sensing, and communications. Considered first is the steady force exerted by traveling waves on scatterers that are smaller than a wavelength. Asymmetric material distributions are found to experience forces that are two orders of magnitude weaker than forces exerted on symmetric distributions. Considered next is the aggregate effect of subwavelength asymmetric piezoelectric scatterers suspended periodically in a fluid. The asymmetry couples the effective acoustic and electrodynamic constitutive relations, which are shown to conserve energy and obey reciprocity. Attention is finally turned to acoustic fields with helicoidal wavefronts that break axisymmetry. Analytical expressions of such waves are obtained for all frequencies, yielding scaling laws and geometric descriptions of the fields. In the absence of asymmetry, the findings of this dissertation simplify to previous results for scattering and diffraction from symmetric objects.